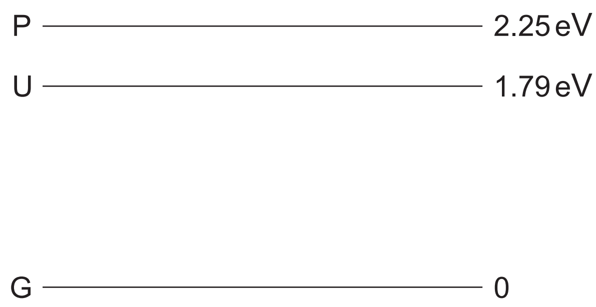
Examiner only

9. A simplified energy level diagram for a three-level laser is given below.



(a) The laser emits light by means of stimulated emission involving levels U and G.

(i) Explain what is meant by stimulated emission involving levels U and G. [2]

....

....

....

(ii) State why there must be more electrons in U than in G for light amplification to occur. [1]

....

....

(b) Calculate the wavelength of the light emitted by stimulated emission. [3]

....

....

....

....

Examiner only

(c) The laser emits light at a power of 0.60 W.

(i) Show that approximately 2×10^{18} photons are emitted per second. [1]

....

....

....

(ii) Calculate the magnitude of the momentum of an emitted photon. [2]

....

....

....

(iii) The light from the laser strikes a shiny surface at right angles. Assuming that the surface reflects all the light, calculate the force exerted on the surface by the light. [2]

....

....

....

....

11

END OF PAPER